Laporan Sementara Elektronika Digital 2

Dasar dasar rangkaian sequensial 2

Anggota Kelompok :

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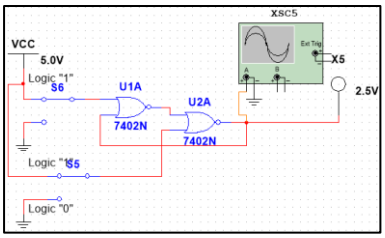
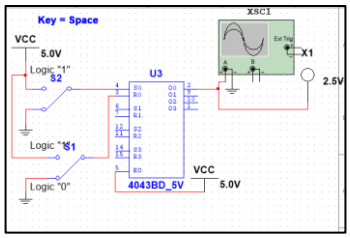
2. Muhammad Slamet Ridho Novianto (2425600050)
3. Arka Pandita Ari Dermawan (2425600057)

A. SR Flip-flop dari gerbang NOR dan NAND

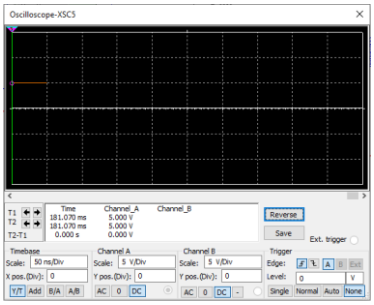
- Gerbang NOR



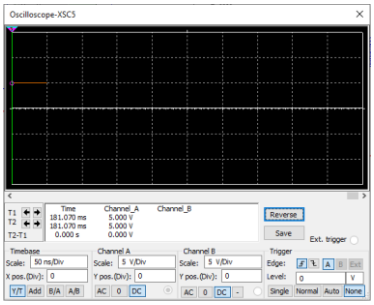
Rangkaian dengan SR-FF NOR



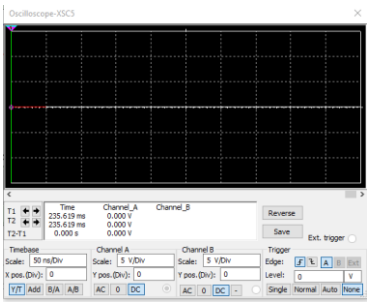
Rangkaian dengan IC SR-FF NOR



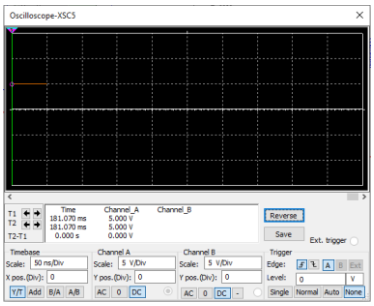
Oscilloscope logic “00”



Oscilloscope logic “01”



Oscilloscope logic “10”

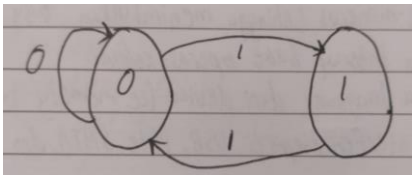


Oscilloscope logic “11”

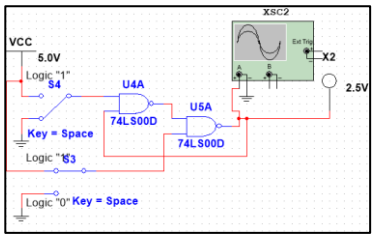
Table PS/NS

S	R	PS(Qn)	NS(QN+1)
0	0	Zn	Zn
0	1	0	1
1	0	1	0
1	1	0	0

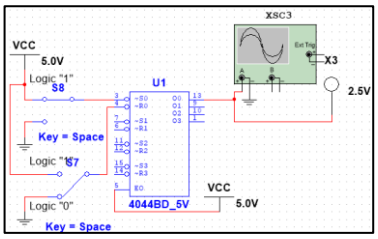
State Diagram (NOR)



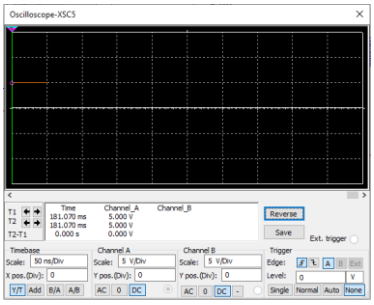
- Gerbang NAND



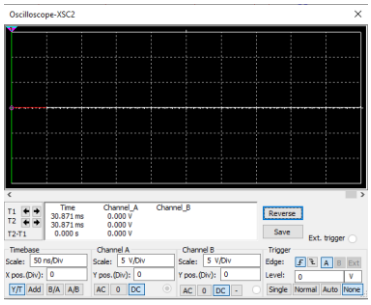
Rangkaian dengan SR-FF NAND



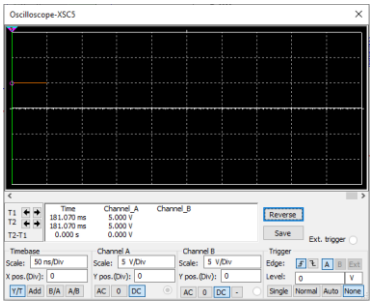
Rangkaian dengan IC SR-FF NAND



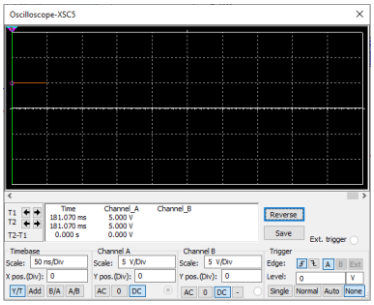
Oscilloscope logic “00”



Onscilloscope logic “01”



Oscilloscope logic “10”

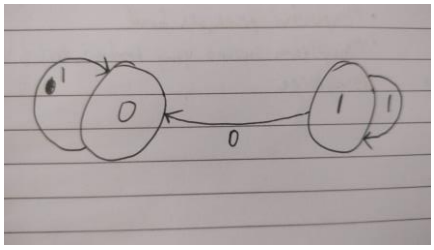


Oscilloscope logic “11”

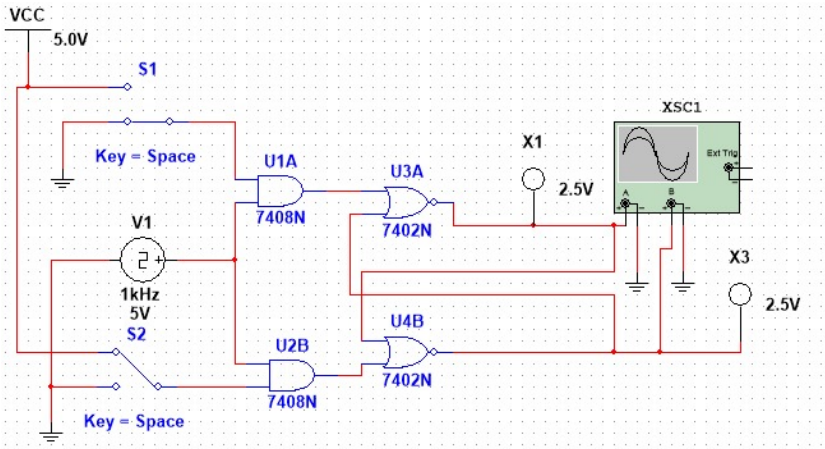
Table PS/NS

S	R	PS(Qn)	NS(QN+1)
1	1	0	0
0	1	1	0
1	0	0	1
0	0	1	1

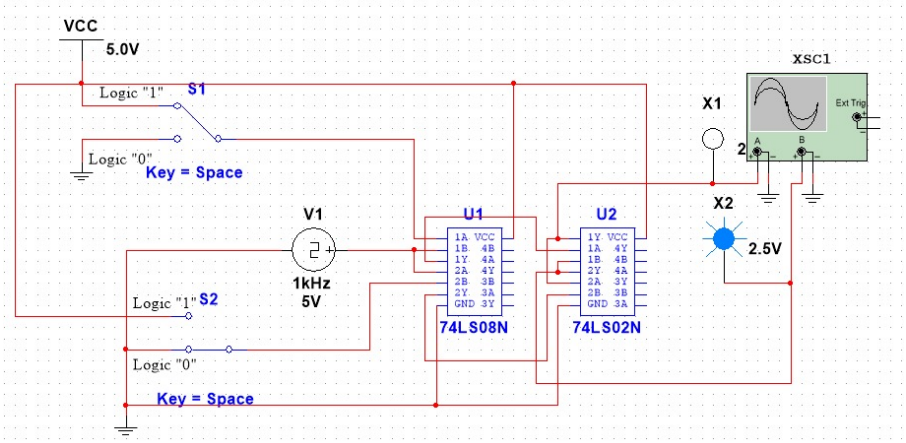
State diagram (NAND)



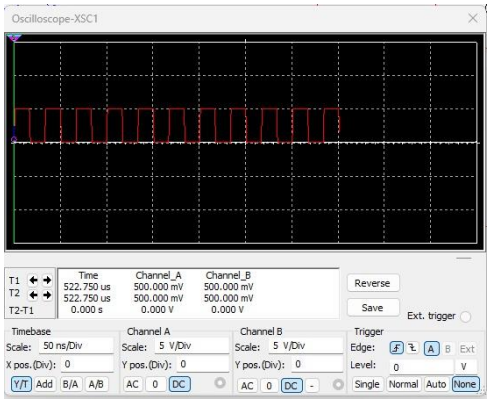
B. SR Flip-flop dengan Clock



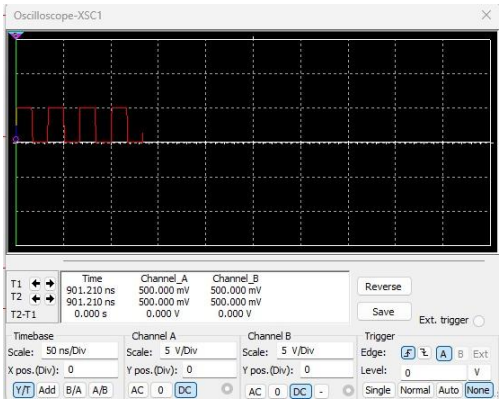
Rangkaian dengan Gate



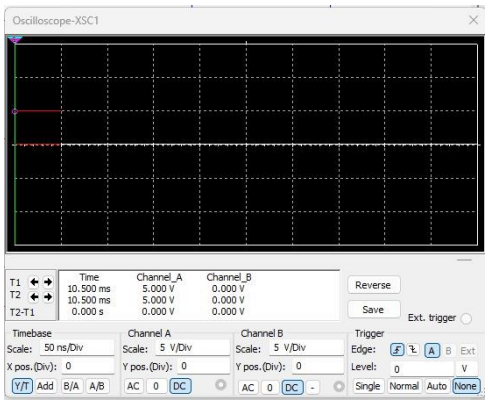
Rangkaian dengan IC



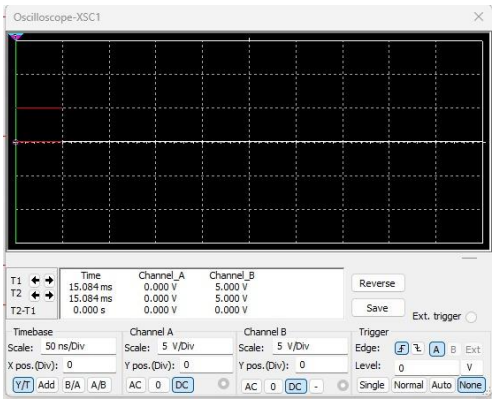
Oscilloscope logic “11”



Oscilloscope logic “00”



Oscilloscope logic “01”

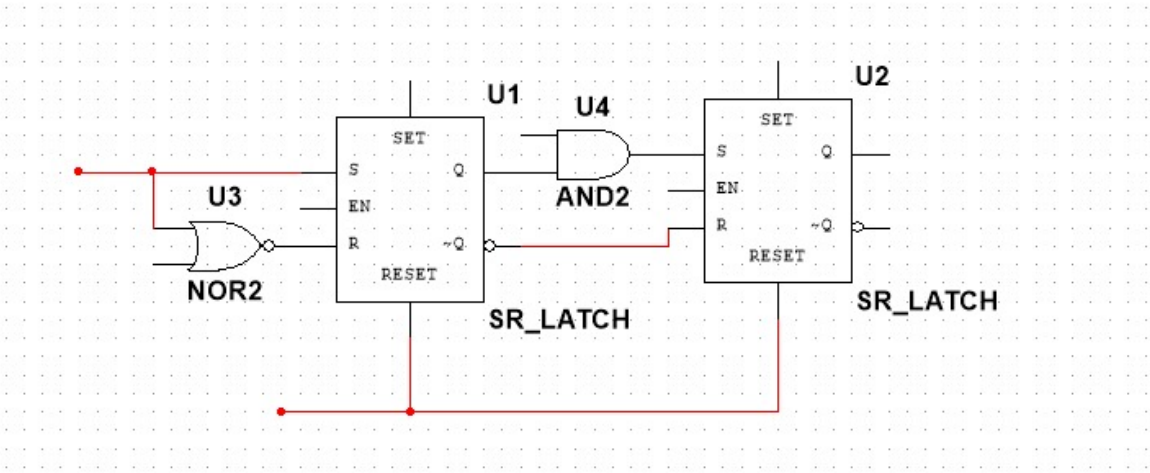


Oscilloscope logic “10”

Tabel PS/NS

C	S	R	PS	NS
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	1	0	0	1
1	1	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	1	0	X
1	1	1	1	X

C. Analisa Rangkaian Sekuensial



Tabel PS/NS

S	R	Q(t)	Q(t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

D. Sintesa Rangkaian Sekuensial (dengan JK-FF)